Niraj Dhakal

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Summary

AI engineer with 16+ months of production experience building LLM-powered systems end-to-end. Shipped multi-user retrieval-augmented chatbot on CPU-constrained infrastructure, including LLM inference optimization, context window engineering, and architectural migration to hosted inference. Strong in Go, Python, and PostgreSQL.

Skills

Languages: Go, Python, TypeScript/JavaScript, SQL (PostgreSQL), C/C++
AI/ML: LLM integration (Ollama, hosted APIs), RAG architecture, prompt engineering, context window optimization, pgvector
Tools & Technologies: Gin, Celery, Selenium, Next.js, React, Prisma
Infrastructure: Docker, PostgreSQL, MongoDB, GIT, REST APIs
Concepts: Multi-user architecture, JWT authentication, async pipelines, agile/scrum

Education

University of Maryland Baltimore County
Bachelor of Science, Computer Science | GPA: 3.667 | May 2026- Cum Laude
Relevant Coursework: Machine Learning, AI/Algorithms, Database Systems, Software Engineering, Malware Analysis

Experience

Software Engineering Intern | Dogwood Gaming | *January 2025 - Present*

- Owned end-to-end design and implementation of LLM-powered chatbot subsystem for an AI marketing platform, including retrieval pipeline, prompt optimization, and production LLM inference deployment.
- Designed retrieval system using PostgreSQL hybrid full-text search and trigram similarity over game documentation and scraped competitor data.
- Debugged and optimized CPU-only LLM inference (Qwen2.5:3B through Ollama) including model quantization migration reducing inference latency from 5+ minutes to 3.5 minutes. Led architectural migration to hosted LLM API resulting in 5 seconds end-to-end query latency.
- Built persistent chat history in PostgresSQL with JSONB storage and group-scoped authorization for chatbot.
- Refactored authentication from LocalStorage to HTTP-only JWT cookies with multi-user authorization, mitigating CSRF and eliminating XSS based token theft across 16+ API endpoints.
- Containerized full stack using Docker. Designed deployment scripts and weekly schema/data backups for rollback.

Scrum Master/Developer | Retreiver Proposal Portal | *Fall 2025*

- Developed a full-stack project portal using **Next.js, Prisma, and MongoDB** that connects UMBC students with faculty-led initiatives.
- Built role-based access control for three distinct user types (students, stakeholders, admins), enabling secure project discovery and collaboration.
- Collaborated with faculty stakeholders to define requirements and iterate on features, resulting in the platform being presented to campus leadership as a potential university-wide solution.
- Managed sprint cycles using **Jira**, including backlog prioritization, task decomposition, and progress tracking to ensure timely task completion.

Teaching Assistant | CMSC 201 | *UMBC Fall 2025*

- Hosted weekly labs for 18 students, teaching students about Python and coding fundamentals.
- Held weekly office hours and graded assignments, providing feedback on code quality and logic.

Projects

LocalStream | Real-time Screen Sharing Application | *https://github.com/Niraj-Dhakall/local-stream*

- Built a peer-to-peer screen sharing platform using WebRTC for media streaming and a Go WebSocket server for signaling, supporting 5+ concurrent cross-device viewers over a local network with HTTPS.
- Implemented thread-safe room management with `sync.RWMutex` to handle concurrent viewer connections and prevent race conditions in shared state.
- Scaled architecture from 1-to-1 to 1-to-many connections, maintaining direct P2P media paths while the server handled signaling only (no media relay).